Sample Mark Scheme: P000310

NCFE Functional Skills Qualification in Mathematics at Level 1 (501/2325/7)

Activity 1		Marks
1A1	**10p or £0.10 (accept 5p each) units needed**	**1**
1A2	**It is 10p more to buy 6 than to buy 7**	**2**
	(£) 2.10 seen OR 210 (p) OR 6 x 0.35 OR 2/7=28.6 or 29p	1
	A correct comparative statement e.g. cheaper to buy 7 The price for 6 must be compared with the price of 7 (don't need actual price difference but needs to be supported by values)	1
1A3	**20 (p)**	**2**
	100/6 + 3 or 6*0.03 + £1 = 1.18/6 or 17(p) seen or 17 + 3 or 16 + 3 seen	1
	Correct answer only (CAO) 20(p) (accept £0.20 but must be shown to 2 decimal places with £)	1
1B1	**(£) 60**	**2**
	9 x 20 /3 OR 9/3 = 3 x 20 OR equivalent (don't accept 0.3 as 1/3)	1
	CAO (£) 60	1

Sample

Activity 1		Marks
1B2	**8 with valid check**	**3**
	1200 (g) or 0.15 (kg)	1
	CAO 8	1
	Check using reverse calculation, for example, 8 x 150, 8 x 0.15	1
1C1	**9.3 (or 9 with 9.3 seen) with valid check**	**3**
	sum of numbers/10	1
	9.3 (accept 9 but only if 9.3 seen)	1
	Check using reverse calculation, for example, 9.3 x 10 = 93 (accept 9 x 10 = 90)	1
1C2	**5**	**1**
	Total marks:	**14**

Activity 2		Marks
2A	**20**	**4**
	5 or 75/15 or 75 x 60 = 4500 (m sq)	1
	4 or 60/15 or 15 x 15 = 225 (m sq)	1
	Follow through (FT) their 5 x their 4 or diagram or Area method with correct values (4500/225)	1
	CAO 20 (bowls)	1

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Activity 2		Marks
2B	**£9.90 OR 990p**	**4**
	270 (cm) seen or 2 (75 + 60) OR 2 (0.75 + 0.6) or using area of sides	1
	plus 60 (cm) or plus 0.6 (m) OR 0.6 (m sq) if using area of top	1
	FT their length x (£)3 or their area x (£)3 (accept unrounded or rounded values x (£)3), don't accept their volume x £3	1
	CAO £9.90 or 990p	1
2C1	**5m^2 units required e.g. 5 m sq or 5 M2 (may be seen in workings box)**	**1**
2C2	**The perimeter of A is (2m) longer than the perimeter of B, with 10 and 12 seen**	**3**
	Plan A 12 (m)	1
	Plan B 10 (m)	1
	FT accurate comparison on 2 perimeters seen (don't accept just 'difference' - need larger/smaller, supported by values) FT on perimeters may include those of equal length e.g. "they are the same"	1
	Total marks:	**12**

Activity 3		Marks
3A1	**3:1**	**2**
	use of 18 and 6 in either order	1
	CAO 3:1, 3 to 1 (don't accept 1:3)	1
3A2	**12 (%)**	**2**

Sample

Activity 3		Marks
	3/25 x 100 OR equivalent	1
	CAO 12 (%)	1
3A3	**Correct chart**	**3**
	appropriate, consistent scale intervals displayed	1
	1 mark for any 1 correct bar, either: 25 green, 15 red or 24 black (these can be awarded if only 0 and 30 are on scale). Apply tolerance of 0.5 sq /2.5 cases	1
	1 mark for all 3 bars correct: 25, 15 and 24	1
3B1	**60 (p) (accept £0.60, must be shown to 2 decimal places and with £)**	**1**
3B2	**92p (accept £0.92), units required**	**2**
	12 (p) seen or 15% of 80 e.g. 80 x 15/100	1
	CAO 92 p (accept £0.92) units required	1
3C1	**1/20**	**2**
	4/80 (accept 80/4, 80/4=20)	1
	CAO 1/20 or 1 in 20 (accept 5%)	1
3C2	**(£)13.68**	**2**

Sample

Activity 3		Marks
	76 x 18 (p) or 76 x 0.18	1
	CAO (£)13.68 (award if (£)13.68 seen in workings but (£)14 at final answer)	1

Total marks:	**14**

Overall marks:	**40**
Pass mark:	**26**

Summary of Skills Standards and Coverage and Range

(Note: where task reference and marks are indicated against a skill standard they can be for any of the associated coverage and range statements)

Skills standards	Total marks	Required weighting	Actual weighting	Coverage and range (can be covered across all skills standards)	Task reference	Marks awarded
Representing **R1** understand practical problems in familiar and unfamiliar contexts and situations, some of which are non-routine **R2** identify and obtain necessary information to tackle the problem **R3** select mathematics in an organised way to find solutions	**14**	**30 - 40 %**	**35%**	**a.** understand and use whole numbers and understand negative numbers in practical contexts	**1A1, 1A2, 1A3, 1B1, 2A, 2A, 3C2**	**7**
				b. add, subtract, multiply and divide whole numbers using a range of strategies		
				c. understand and use equivalencies between common fractions, decimals and percentages	**3A2, 3B1, 3B2, 3B2**	**4**
				d. add and subtract decimals up to two decimal places		
				e. solve simple problems using ratio where one number is a multiple of the other	**1B1, 1B2, 3A1**	**3**
				f. use simple formulae expressed in words for one- or two-step operations		

Sample

Analysing **A1** apply mathematics in an organised way to find solutions to straightforward practical problems for different purposes **A2** use appropriate checking procedures at each stage	**13**	**30 - 40%**	**32.5%**	**g.** solve problems requiring calculation with common measures, including money, time, length, weight, capacity and temperature	**1A2, 1A3, 2B**	**3**
				h. convert units of measure in the same system	**1B2, 1B2, 2A, 2B, 2B, 2B, 2C1, 2C2, 2C2, 2C2, 2C2, 3C2**	**11**
				i. work out areas and perimeters in practical situations		
				j. construct geometric diagrams, models and shapes		
Interpreting **I1** interpret and communicate solutions to practical problems, drawing simple conclusions and giving explanations	**13**	**30 - 40%**	**32.5%**	**k.** extract and interpret information from tables, diagrams, charts and graphs	**2A, 3A1, 3A2, 3A3. 3A3, 3A3,**	**6**
				l. collect and record discrete data and organise and represent information in different ways		
				m. find mean and range	**1C1, 1C1, 1C1, 1C2**	**4**
				n. use data to assess the likelihood of an outcome	**3C1, 3C1**	**2**
Total marks:	**40**					**40**

Question Type	
Open:	**34 (85%)**
Closed:	**6 (15%)**

Sample